

C++ Documentation: Strings, Structs, Functions, and Enums

1. Strings

C++ provides the `std::string` class for text manipulation.

- `length()/size()`: Return the number of characters.

Example:

```
std::string s = "Hello";
size_t len = s.length(); // len = 5
```

- `substr(pos, count)`: Extract substring.

Example:

```
std::string sub = s.substr(1, 3); // sub = "ell"
```

- `find(str)`: Find first occurrence.

Example:

```
size_t pos = s.find("ll"); // pos = 2
```

- `count (algorithm)`: Count occurrences of a character.

Example:

```
int cnt = std::count(s.begin(), s.end(), 'l'); // cnt = 2
```

- `toupper/tolower (ctype)`: Convert character case.

Example:

```
char c = std::toupper(s[0]); // c = 'H'
```

2. Structs

Define custom types with multiple fields.

```
/**
 * @brief Represents a 2D point.
 */
struct Point {
    int x, y;
    // Equality operator
    bool operator==(const Point &other) const {
        return x == other.x && y == other.y;
    }
};

std::pair<int,int> p = {1, 2};
std::tuple<int,int,int> t = {1, 2, 3};
// Unlike struct, std::pair and std::tuple do not have named fields.
```

3. Functions

Functions encapsulate reusable logic.

```
/**
 * @brief Computes factorial recursively.
 * @param n Non-negative integer
 * @return n! (factorial)
 */
int factorial(int n) {
```